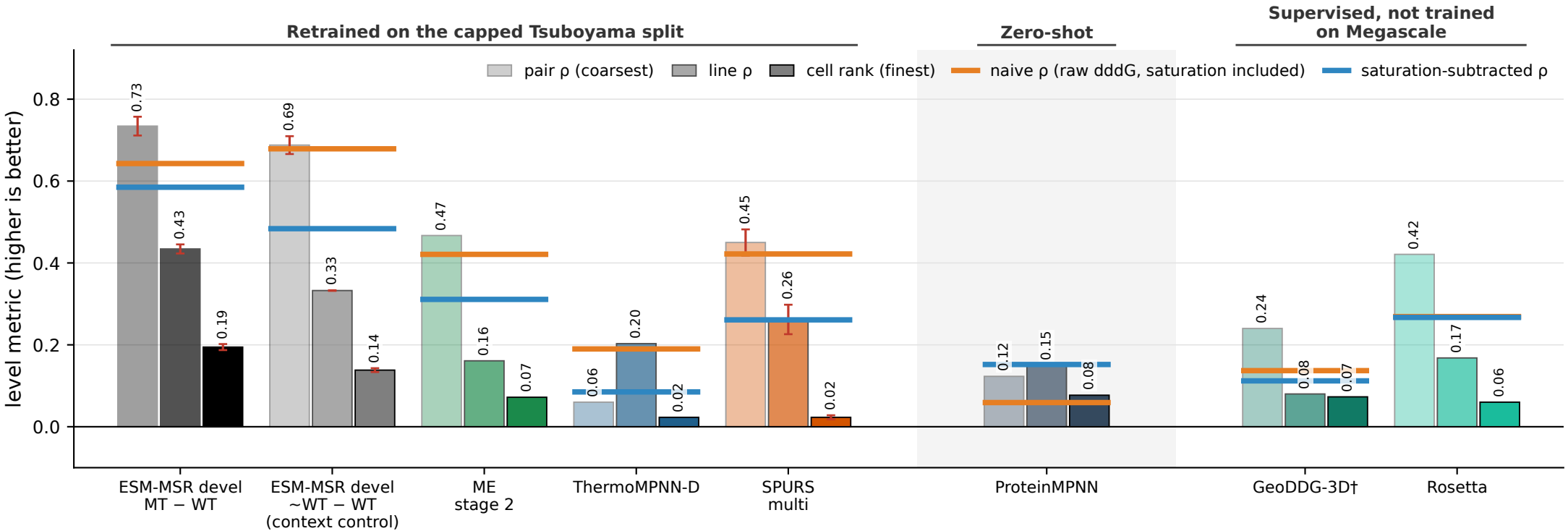
Validation: epistasis predicted level by level, 8,494 doubles with both singles measured, 24 position pairs



† trained on S8754 (ProThermDB + ThermoMutDB), not on Megascale. ESM3-small-open and ProSST are not in the epistasis tables; ESM3ΔG is left out (overlap with its MGnify fine-tune).
MT − WT: the epistasis head. ~WT − WT: the WT adapter read on the mutated sequence, a control for what the backbone and single-mutant training know; both scored by the epistasis suite (3 devel seeds).
Error bars: sd over 3 seeds (ESM-MSR devel, SPURS multi); other rows are single checkpoints or public models, with the benchmark document's numbers (tables 2.1 / 2.2).